05-04-2025

Agenda

operators
Control Flow
User input
strings

Arithmetic operators ✓

Assignment operators ✓

→ =

+= ⇒ a = a + 1
-= ⇒ a = a - 1
/= ⇒ a = a / 1
//= ⇒ a = a // 1
%=
**=

Example with 1

a = 2 → a += 1

a = a + 1
a = 2 + 1
a = 3

a += 1

P. increment a by 1

a = 2
a = a + 1 / a += 1
a = 3

a = 10 , 10

→ a = a + 10
a += 10

Comparison operator → Boolean

a = 2 , b = 3

(a) == (b)

↓

Boolean

==	Equal
>	greater than
<	Smaller than
>=	Greathan than or equal to
<=	Smaller than or equal to
!=	Not equal to

a = 2 , b = 3	a = 100 , b = 100
a == b → False	True
a > b → False	False
a < b → True	False
a >= b → False	True
a <= b → True	True
a != b → True	False

logical operator

True → 1

False → 0

Binary language

0, 1

00 - (1)
11 - (2)
01 - (3)
10 - (4)

Truth table

Operator	A	B	Answer / output
And	0	0	0
	1	1	1
	0	1	0
	1	0	0

And → True, when both are True else False

Operator	A	B	Answer / output
OR	0	0	0
	1	1	1
	0	1	1
	1	0	1

OR → if any one is True, True, if both are False, False

Operator	A	Answer / output
→ Not	0	1
	1	0

0, 1 → ②

Operator	A	B	Answer / output
XOR	0	0	0
	1	1	0
	0	1	1
	1	0	1

XOR → only when one of it is true

Prob

Q: Find the case where one is say, such & another must

XOR

A	B
1	1
0	1
1	0

Test

True → False

False → True

Not

value → Not → opposite

→ 0 != 0 → True / False

0 → 1

1 → 0

XOR → only where one is true, it is true other wise false

Bit wise

Binary numbers

4	2	1	
0	1	1	→ 3
0	0	1	→ 1

&

|

→ ^

~

```
  011
- 001
------
& → 001
```

```
  011
  001
------
| 011
```

```
  011
  001
------
^ 010
```

~ 011 → 3

100 → 4

logical operator

a = 2 , b = 3 , c = 5 , d = 8

→ → (a > b) and (c < d) → False

→ (False) and (True)

False (and) True

0 1

→ False

→ a = 2

→ b = 2

→ c = a + b

→ printf(c)

1 ← 2

2 ← 1

3 ← 4

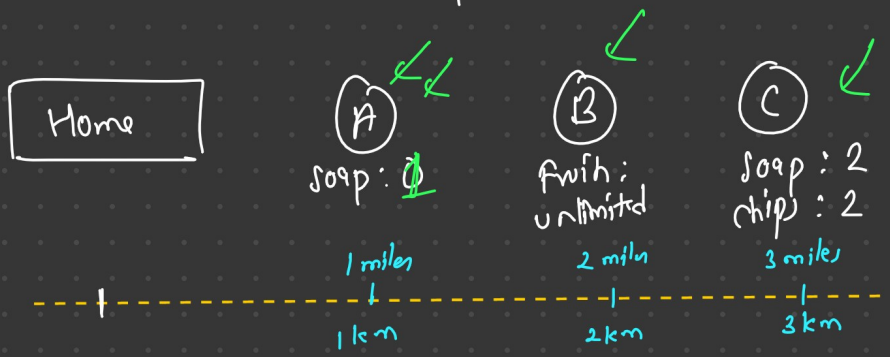
4 ← 3

Control flow → control the executing of code

Conditional Statement

Task →

Buy few items:
- chips
- oranges
- soap



start

A → do you have soap?
if you have soap, i want one
Buy
else
go to the next store →

B → do you have fruit?
if yes:
buy → buy
else
move to next store

C do you have soap & chips:
if yes
Buy → buy, go home
else:
return home

if else

⟹ if (condition) :

do something

⟶ else :

go to next step do otherwise

Zoom

name = " "

```
if name == "monal":
    print("welcome instructor")
else:
    print("welcome attendee")
```

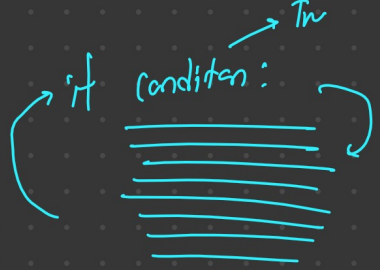
① name = "monal"

② if name == "monal":
    print("Welcome instructor")
else :
    print("Welcome attendee")

indentation

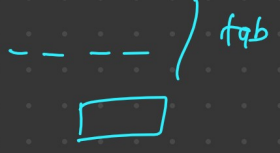
↓

area, block



press tab → 4 space bar





tab / 1 space / 3 space

Syntax

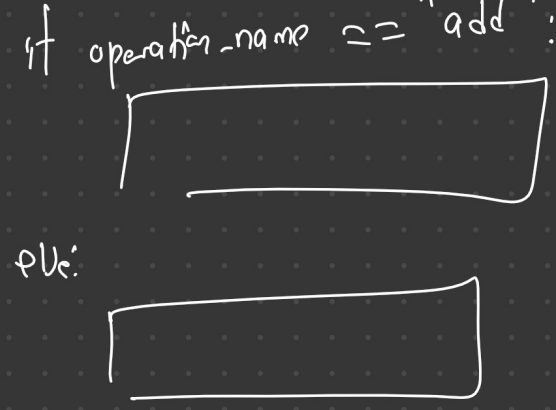
```
if condition :
    do some thing
else :
    otherwise
```



problem statement

if i have to build → calculator

- → add
- → subtract
- → multiplication
- → division



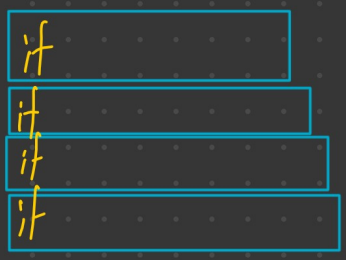
① if
else

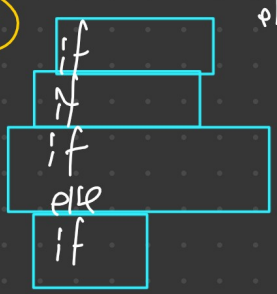
② if — ①
elif — n
else — ①

③ if

④ if
elif
elif
elif

⑤ if
elif
elif
elif
elif
else





```
operation_name = "multiply"

if operation_name == "add":
    print("We are adding 2 numbers")
elif operation_name == "subtract":
    print("We are subtracti 2 numbers")
else:
    print("We are not adding anything")
```

"add"

True

else will run if all of them if false

run

```
operation_name = "add"

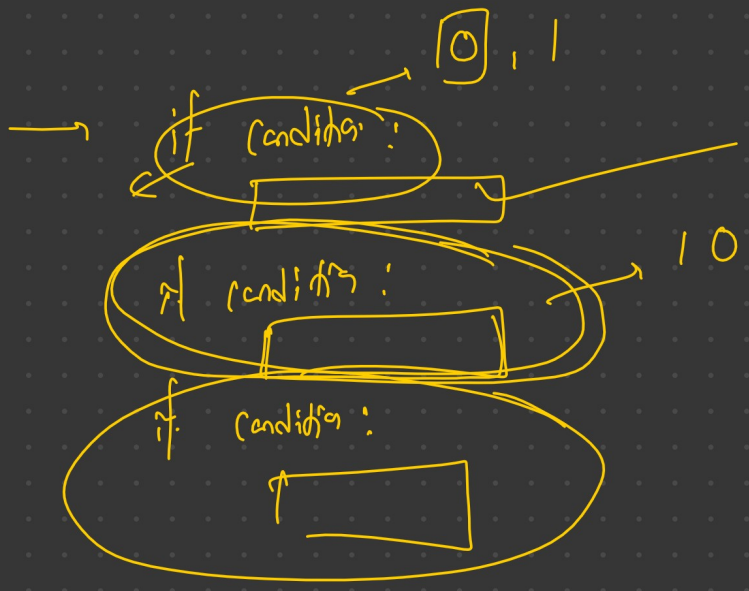
if operation_name == "add":
    print("add")
if operation_name == "subtract":
    print("Subtract")
if operation_name == "multiple":
    print("multiply")
```

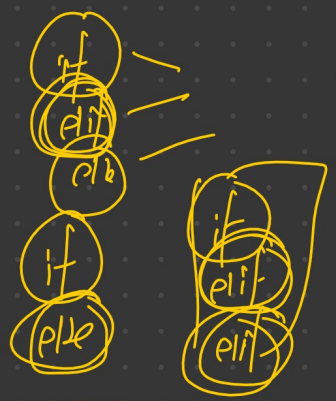
True

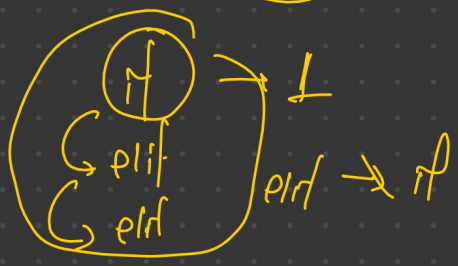
3 times

min / max → 3

max → 3, 1







else → ✗

else →

→ money = input()
→ time = input()

assignment

check whether you have money or not,
if you have money,

check if you have >200
    check if time is less than 7
        go to theme park
    else
        go home
if you have <200:
    check if time > 7
        go to icecream parlor
    else
        go to beach

Consider time as PM

a = 2
b = 3
→ static

a = input()

input
Name